# MidNet: An Anchor-and-Angle-Free Detector for Oriented Ship Detection in Aerial Images

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***Abstract*— Ship detection in aerial images remains an active yet challenging task due to its arbitrary object orientation and various aspect ratios from the bird's-eye perspective. Most existing oriented objection detection methods rely on angular prediction or predefined anchor boxes, making these methods highly sensitive to unstable angular regression and excessive hyperparameter setting. To address these issues, we replace the angular-based object encoding with an anchor-and-angle-free paradigm and propose a novel detector deploying a center and four midpoints for encoding each oriented object, namely, MidNet. Moreover, MidNet designs a novel symmetrical deformable convolution for enhancing the features of midpoints; then, the center and midpoints for an identical ship are adaptively matched by predicting corresponding centripetal shift and matching radius. Finally, a concise analytical geometry algorithm is proposed to calculate the ship orientation and refine the keypoints stepwisely for building precise oriented bounding boxes. On two public ship detection datasets, HRSC2016 and FGSD2021, MidNet outperforms the state-of-the-art detectors by achieving APs of 90.52% and 86.50%.**

***Index Terms*— Aerial images, anchor-and-angle-free detector, deep learning, ship detection.**

## I. Introduction

AMONG a variety of object detection tasks, ship detection in aerial images has been drawing an increasing attention for homeland security, emergency rescue, and maritime situational awareness. Due to arbitrary orientation and large aspect ratio of the ship from a bird's-eye view, horizontal bounding boxes are not adequate for describing its location information. In recent years, oriented ship detection has attracted extensive academic attention [1], [2], [3].

At present, a widely adopted paradigm for handling oriented objects is to attach an extra branch to the boundary box regression network for predicting rotation angles. Fig. 1(a) and (b) illustrates the two basic coding manners for angular prediction in anchor-based and anchor-free forms. For anchor-based methods, additional anchors with multiple angles need to be tiled for each point on the feature maps [see Fig. 1(a)], while anchor-free methods need to build a regression branch to learn the rotation angles of objects [see Fig. 1(b)]. Although there will be varieties in the details of coding manners, the angular-based paradigm limits their further improvement because of the following two reasons.

1) *The Boundary Discontinuity of Regression:* Owing to the periodicity of angles and the exchangeability of edges, a small prediction deviation may bring a great loss increase under the boundary condition. For example, the 1° and 365° are very close in practice, but they are far from the absolute value. The calculated error does not meet the actual expectation.

2) *High Sensitivity to Angular Prediction Errors:* A slight angular prediction error will cause huge intersection of union (IoU) reductions, especially for objects with large areas or large aspect ratios.

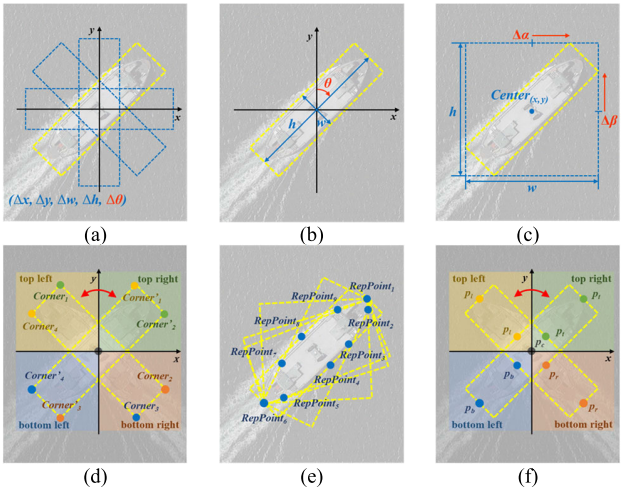



Fig. 1. Coding manners of oriented object. (a) and (b) Two encoding manners based on angle prediction, which are implemented in anchor-based and anchor-free form, respectively. (c) Angle-free prediction method, but this method still has the limitation of anchor mechanism. (d)–(f) Keypoint detectors. (d) Corner case. As the angle changes, the top left corner (the yellow point) may fall in the top right quadrant. The orientation definition of corners is ambiguous for oriented objects. (e) RepPoint case, where the yellow represents the potential bounding boxes, and these bounding boxes are not unique for an oriented object. (f) Proposed midpoint case. No matter how the angle is rotated, the orientation of the midpoint corresponds to that of the quadrant. It is beneficial to stable learning of features.

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To alleviate unstable angle learning, there are different attempts to transform angular prediction into other ways. Some researchers locate the intersection of the object and its external rectangle to rotate a horizontal bounding box into an oriented one. Gliding Vertex [4] regresses four length ratios representing the relative gliding offset on each corresponding side, shown in Fig. 1(c). Oriented region-CNN (R-CNN) [5] predicts the offsets from the middle position of the sides to the intersection. However, these methods require predefined horizontal boxes as the starting status, which is usually implemented in the form of anchors. Thus, they inevitably cause excessive manually adjusted hyperparameters, which limits the generalization capability for novel scenarios and consumes excessive computational cost.

Keypoint-based detector [6], [7], [8], [9], [10] has attracted extensive academic attention in horizontal object detection by providing an alternative method for encoding the object location using representative keypoints, and it provides a new perspective for angle transformation in oriented object detection. However, there are still some challenges in directly applying the horizontal detection method to oriented objects. First, the definition of keypoints in most existing detectors is invalid for oriented objects. Take the corner-based detectors as an instance, the corners can always be defined by their relative positions to the center point in the horizontal box. For example, the "top left" corner always appears in the top left of the object. However, due to the arbitrariness of the object orientation, the distribution of corners is not universal. As shown in Fig. 1(d), when the ship orientations are in two different directions, the top right corner represented by green point is not always in the green area at the top right of the object. Such inconsistency will not only cause the confusion in the definition of corners but also instability in the process of feature learning. Another challenge in using keypoints for encoding rotated objects lies in the uncertainty of the side direction of objects. For the horizontal object, there are always two vertical and horizontal sides in the bounding box. Therefore, a unique external bounding box can be easily constructed according to the keypoint estimation, such as RepPoint [6] and ExtremeNet [7]. On the contrary, for the oriented object, the uncertainty of the side direction makes the external rectangle of the keypoints nonunique, as shown in Fig. 1(e). Thus, it is difficult to build a precise bounding box without angle information. Zhang et al. [1] proposed an oriented ship detector, which locates the object through the head and center point. However, this method requires additional prior information of the head points and increases the cost of annotation.

To conquer these issues, we propose a keypoint-based representation for oriented ship detection and design a novel anchor-and-angle-free detector, MidNet. The midpoints of the four edges and the center point of the object are utilized to encode the bounding box, as shown in Fig. 1(f). This representation avoids the definition confusion of the oriented keypoints. According to this idea, MidNet predicts the center point and its maximum inscribed circle radius, as well as the midpoints and their centripetal vectors. For the midpoints, a novel deformable convolution is designed to enhance the features by using the prior information of ship's structural symmetry, especially for trailing or obscured ships. After obtaining the keypoint estimation, the keypoints are suppressed and matched adaptively according to the learned radius and centripetal shift. Then, the oriented bounding box is built by the refined keypoints through an analytical geometry method.

The contributions of this article can be summarized as follows.

1) A novel oriented object representation based on midpoints is proposed, which not only makes the framework avert the instable angular regression and excessive hyperparameter setting but also eliminates the confusion in the keypoint definition and learning process.
2) An anchor-and-angle-free detector, MidNet, is proposed for ship detection in aerial images. Moreover, a symmetrical deformable convolution is customized for locating the ship keypoints, where the feature enhancement of the symmetrical midpoints and center is realized with only once offset prediction.
3) A concise analytical geometry method is designed to refine and construct keypoints into precise oriented rectangular boxes, and the increase of parameters and extra time consumption is negligible.
4) Extensive experiments and visual analysis on different datasets prove the effectiveness of our technique. The proposed method achieves 90.52% and 86.50% mAPs on HRSC2016 and FGSD2021, which outperforms the state-of-the-art methods.

## II. Related Work

### A. Oriented Object Detectors

Oriented detectors attract extensive academic attention in remote sensing and scene text images. Most of the existing methods are derived from the anchor-based multistage detectors. In the initial exploration, rotate region proposed network (RRPN) [11] obtained the rotated region proposal by tiling rotated anchors, and then, RoI-Trans [12] transformed a horizontal region of interest (RoI) into a rotated RoI (RRoI) through RRoI learner and RRoI wrapping. In ReDet [13], Han et al. introduced the rotation equivariance features, and a new RiRoI Align was designed for rotation-invariant features extraction in the introduced rotation equivariant.

The single-stage network has also attracted academic attention due to its flexible structure. In $R^3$Det [14], Yang et al. proposed a refined single-stage rotated detector to solve the problem of feature misalignment. In $S^2$A-Net [15], Han et al. proposed a new alignment convolution, which alleviates the shifting in the process of alignment between axis-aligned convolutional features and oriented objects. However, like the horizontal detectors, these anchor-based methods suffer from excessive hyperparameters, high computational cost, and low adaptability.

Meanwhile, the unstable angle prediction is another central issue in oriented object detection. SCRDet [16] and RSDet [17] tried to alleviate the boundary discontinuity through the loss function. In circular smooth label (CSL) [18] and densely coded labels (DCL) [19], angle regression was

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considered as a classification task in order to avert the boundary location problem. In Gliding Vertex [4] and Oriented R-CNN [5], the frameworks construct the oriented bounding box by locating the intersection of the object and its external rectangle. However, these methods either fail to completely avoid angle prediction or require predefined anchors, which limits the further improvement of detection performance. In $O^2$-DNet [20], the oriented object is represented by a pair of middle lines in an anchor-free fashion, and each middle line is defined as the line connecting the midpoints of two opposite sides of an oriented bounding box, but the implicit regression learning process and neglect of rotation feature extraction limit its performance to a certain extent.

### B. Keypoint-Based Detectors

For the horizontal object detection, anchor-based detectors suffer from sample imbalance, redundant hyperparameters, and limited generalization ability. The keypoint-based methods have attracted extensive academic attention due to their flexibility and adaptability. These approaches predict the object keypoints to generate the bounding boxes.

Law and Deng [8] used two corner points to encode objects. The proposed CornerNet predicts the heatmap and offset of the corners, so as to locate and fine-tune the keypoints. In addition, corner pooling is proposed to aggregate the salient features of the object. On the basis of CornerNet, CenterNet [9] predicted the center point additionally. In terms of the keypoint matching strategy, the above two methods predict and compare the similarity of embedding vectors to assist keypoint matching. However, for objects with similar appearance features, the above two methods may produce similar embedding vectors, resulting in incorrect keypoint matching. To avoid such implicit learning, CentripetalNet [10] predicted the centripetal shift of the corner points and matched the corners whose shifted results are aligned. Except for corners, ExtremeNet [7] uses extreme points on the four edges to represent the object, but this method needs more accurate annotation. In the postprocess, ExtremeNet traverses all possible keypoint combinations and determines which ones are valid, so as to complete the keypoints matching. RepPoint [6] proposed a new object representation, which generates a certain number of representative points for each object. It takes the center point as initialization and predicts the outward offset of each representative point to the boundary direction according to the deformable convolution [21]. Then, it obtains the final detection results through the offset keypoint.

A series of keypoint-based methods have made remarkable achievements. However, due to the challenges of extracting rotation features and encoding the oriented bounding box, these methods have received little attention in the oriented object detection.

### C. Oriented Ship Detectors

Ship detection receives additional attention in the remote sensing data analysis due to its unique importance in rescue and military. Liu et al. [22] developed the HRSC2016 dataset in 2017, which is one of the fundamental benchmarks in this field. FGSD2021 dataset is released by Zhang et al. [1], which has fixed ground sample distance and larger imaging range.

As early attempts, Liu et al. [23] and Wang and Li [24] focused on improving the horizontal RPN and ROI pooling to obtain high-quality rotation proposals. However, these methods lack deeper reflection of rotate feature learning. In rotational Libra R-CNN [25], Guo et al. proposed a balanced feature pyramid to balance the feature of different strides and semantic levels. In [26], a bidirectional recurrent neural network fusing self-attention mechanism was designed. Meanwhile, single-stage ship detectors have also been studied. Zhu et al. [27] realized the oriented ship detection by adding angle regression branches on RetinaNet [28], and the interpolation operation was introduced to align the rotated object features.

Given the limitations of anchor mechanism, some anchor-free methods are proposed. AR2Det [3], GRS-Det [2], and CPS-Det [29] directly generated the oriented boxes by estimating the center points, sizes, and rotation angles of the ships. These methods concentrated on modifying particular modules on the original paradigm to adapt rotation feature learning rather than taking advantage of the prior information of ship structure to make further exploration in the customized representation, leading them failed to averting the instability of angle learning thoroughly. CHPDet [1] determines the orientation of the objects by locating the head points and center points; however, this method requires additional annotation cost.

## III. Proposed Method

MidNet utilizes five keypoints $\{p_c, p_l, p_t, p_r, p_b\}$ to represent an object, where $p_c$ refers to the center point of the object and $\{p_l, p_t, p_r, p_b\}$ are the midpoints on the four sides. We define the point located in the top left quadrant or directly left to the center point as the left midpoint $p_l$ and define the others $(p_t, p_r, p_b)$ in the clockwise order. Compared with other keypoint definition methods, the midpoint of the boundary has the following three significant advantages.

1) Regardless of the ship orientation, the midpoints always fall in the four quadrants (top left, top right, bottom right, and bottom left) uniformly. The generality of midpoint distribution makes the model to exploit the information inside the ships in a more definite way.
2) The head of the ships is usually designed as a sharp shape to reduce the resistance. Compared with corner points, the midpoints are more likely to fall inside the ships. Thus, the midpoint representation contains more salient features of the ships.
3) The midpoint coordinates are easily obtained by the general oriented object annotation way, which does not need any additional annotation cost.

Among four midpoints, a pair of symmetric midpoints form a group. As shown in Fig. 2, MidNet constructs three branches, one of which is used to predict the center and its matching radius, and the other two branches of the same structure predict the midpoint groups and centripetal shifts. In the midpoint branch, a symmetrical deformable convolution is designed

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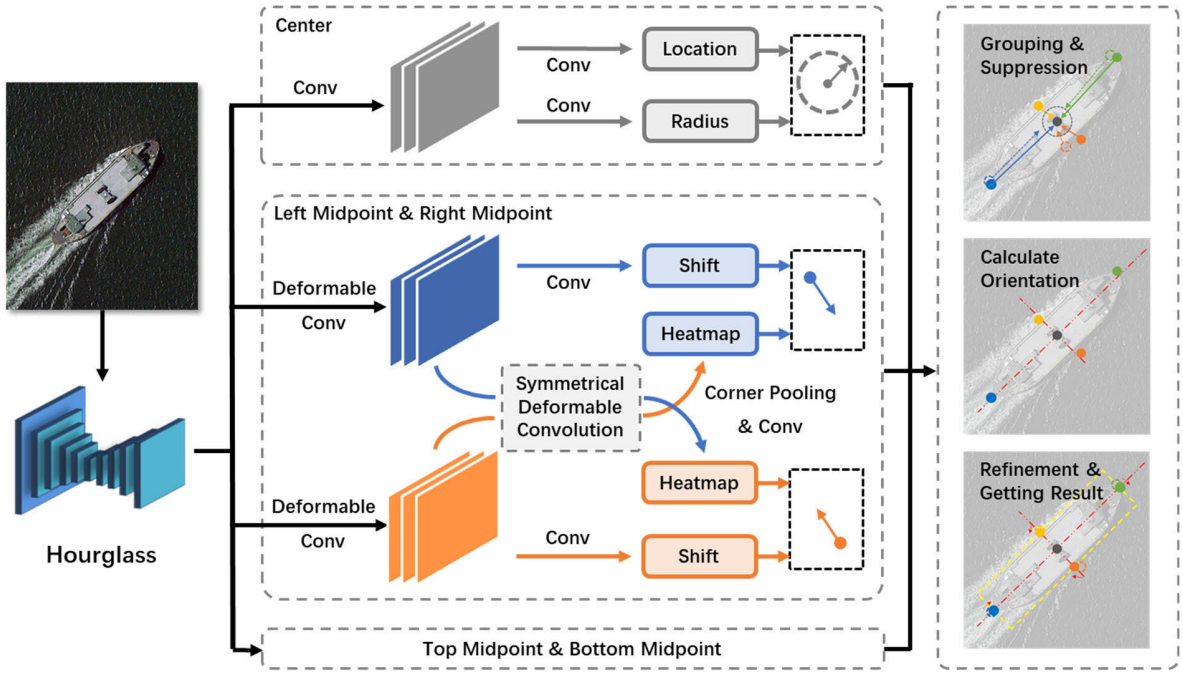



Fig. 2. Overall framework of MidNet. MidNet predicts five keypoints (one center and four midpoints) for ascribing an oriented boundary box. For a pair of symmetric midpoints, such as top left and bottom left midpoints, MidNet predicts the centripetal shift and jointly learns the enhanced features by the symmetrical deformable convolution. For the center, its matching radius is also predicted. Combining the keypoint, radius, and centripetal shift, a series of analytical geometry postprocessing is used to group, suppress, and refine the keypoints. Finally, the keypoints are transformed into oriented bounding boxes.

to promote the feature mining of symmetrical midpoints. After obtaining the keypoint estimation, all the midpoints and centers are matched and suppressed according to the matching radius and centripetal shift. Then, the orientation of the object is precisely calculated under the criterion of minimum confidence-sensitive offset, and the position of the keypoints is refined according to the axisymmetric characteristics of ships. Finally, the refined keypoints are utilized to construct accurate oriented bounding boxes.

### A. Keypoint Estimation

*1) Midpoint Prediction:* In the midpoint branch, a pair of symmetric midpoints are predicted jointly. For each midpoint, it is described by its coordinate, confidence score, and centripetal shift, which are used for positioning, classification, and grouping, respectively. Inspired by CentripetalNet, the centripetal shift $\mathrm{cs}_i$ is defined by the relative position of a midpoint to the center

$$\mathrm{cs}_i = (|x_i - x_c|, |y_i - y_c|) \quad \forall i \in \{l, t, r, b\} \tag{1}$$

where $(x_i, y_i)$ denotes the coordinates of the midpoint in four directions, and $(x_c, y_c)$ refers to the coordinates of the center. For a pair of symmetric midpoints, the symmetrical deformable convolution and corresponding corner pooling are constructed to aggregate the salient features of the object. It should be noted that benefit from the generality of the midpoint distribution, the sign of the centripetal shift is fixed for each midpoint, and only the absolute value of centripetal shift needs to be predicted, which eases the regression process.

In the training phase, the coordinates and confidence scores of midpoints are estimated following the CornerNet pipeline. The focal loss [28] is calculated for supervision

$$L^i_{\mathrm{mid}} = -\frac{1}{N}\sum_j \begin{cases} \left(1-\widehat{h}_j\right)^{\alpha}\log\left(\widehat{h}_j\right), & \text{if } h_j = 1 \\ \left(1-h_j\right)^{\beta}\widehat{h}_j^{\alpha}\log\left(1-\widehat{h}_j\right), & \text{otherwise} \end{cases} \tag{2}$$

where $h_j$ and $\widehat{h}_j$ are the rendered Gaussian map and the predicted heatmap, respectively, $j$ indexes the pixel on the feature map, $N$ is the number of objects, and $\alpha$ and $\beta$ are the weight coefficients. Then, the overall midpoint positioning loss is

$$L_{\mathrm{mid}} = L^l_{\mathrm{mid}} + L^t_{\mathrm{mid}} + L^r_{\mathrm{mid}} + L^b_{\mathrm{mid}}. \tag{3}$$

The centripetal shift is learned with the smooth $L_1$ loss

$$L^i_{\mathrm{cs}} = \frac{1}{N}\sum_{k=1}^{N} \begin{cases} 0.5\left(\mathrm{cs}_k - \widehat{\mathrm{cs}_k}\right)^2, & |\mathrm{cs}_k - \widehat{\mathrm{cs}_k}| < 1 \\ |\mathrm{cs}_k - \widehat{\mathrm{cs}_k}| - 0.5, & \text{otherwise} \end{cases} \tag{4}$$

where $\widehat{\mathrm{cs}_k}$ refers the predicted centripetal shift of the $k$th object, and $\mathrm{cs}_k$ is the ground truth.

Similarly

$$L_{\mathrm{cs}} = L^l_{\mathrm{cs}} + L^t_{\mathrm{cs}} + L^r_{\mathrm{cs}} + L^b_{\mathrm{cs}}. \tag{5}$$

For midpoints, the centripetal shift prediction is category-agnostic but the heatmap specific.

*2) Center Point Prediction:* In aerial images, the sizes of ships range from 300 × 300 pixels (aircraft carriers) to 10 × 10 pixels (yachts). Meanwhile, many ships are densely arranged side by side in the ports. Due to inevitable shift prediction error and inflexible manually set matching range, some existing keypoint matching methods, such as CentripetalNet and CenterNet, would cause incorrect keypoint matching for

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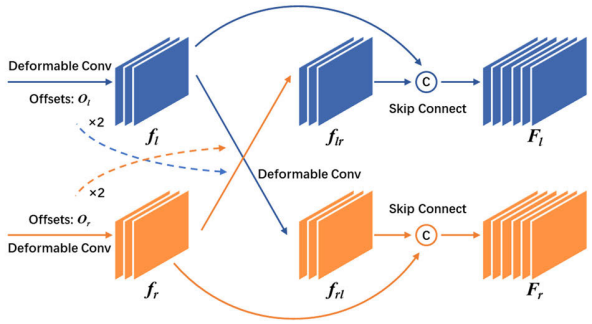



Fig. 3. Structure of symmetrical deformable convolution.

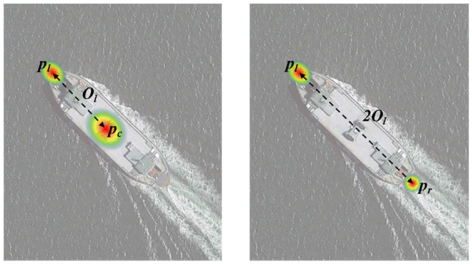



Fig. 4. Spatial symmetrical relation between a group of keypoints for ship objects.

ship detection. In MidNet, the center point and its matching radius $r$ are incorporated to match the midpoints adaptively, where $r$ is defined by the maximum inscribed circle radius and calculated by the distance from the center point to the nearest midpoint

$$r = \lambda \cdot \min(\|(x_c - x_i, y_c - y_i)\|_2 \ \forall i \in \{l, t, r, b\}) \quad (6)$$

which $\lambda$ is the scaling factor.

In the training phase, the center estimation loss $L_{\text{cen}}$ and matching radius loss $L_r$ is calculated in the form of the focal loss and smooth $L_1$ loss, respectively, just like (2) and (4).

### B. Symmetrical Deformable Convolution

The keypoints of ships are often disturbed in a complex port background or the tailed waves in the water, which causes the difficulties in feature extracting of keypoints. To conquer this issue, a novel symmetrical deformable convolution is designed to exploit the features of each other through the mutually symmetrical midpoints. Generally, the feature maps from the backbone are expected to contain salient features of the central regions of the object, and the feature maps in midpoint branch have significant responses to midpoints. Due to the structural characteristics of the ship, the offset from the midpoint to the center is half of that from the midpoint to the symmetrical one. Based on this observation, two deformable convolutions from the same midpoint branch could share the offset to realize the mutual feature perception of symmetric midpoint learning.

The structure of symmetric deformable convolution is shown in Fig. 3; taking $p_l$ and $p_r$ as an example, $O_l$ refers to the ideal offset from the center $p_c$ to $p_l$; thus, twice $O_l$ should equal to the offset from $p_l$ to $p_r$. Therefore, by implementing the deformable convolution with $2{\cdot}O_l$ on $f_l$, $f_{lr}$ could perceive the features around $p_r$ (as shown in Fig. 4). In this case, only once offset prediction is needed to realize the feature extracting and mutual aggregation of symmetrical midpoints.

### C. Building the Precise Oriented Bounding Box

*1) Adaptive Keypoint Matching and Suppressing Scheme:* First, the estimated keypoints are matched and suppressed according to the shifted positions and confidence scores. For each midpoint, the shifted position $(\widehat{x_i'}, \widehat{y_i'})$ is calculated by its corresponding shifts. A midpoint is matched with its closet center if $(\widehat{x_i'}, \widehat{y_i'})$ falls in the matching region, which is defined by the center coordinate and matching radius. After traversing the midpoint, a center is considered valid when at least one midpoint from each direction matches it. In this case, if more than one midpoint from the same direction falls in a valid center, only the one with the highest confidence score will be retained, while the others are suppressed. Compared with manually set matching schemes in CentripetalNet and ExtremeNet, the proposed matching scheme can adjust the matching region adaptively according to different sized objects, which improves the detection performance for crowded and different size ships.

*2) Orientation Calculation and Keypoint Refinement:* For keypoint-based detectors, the estimation of keypoint location is prone to deviations due to the complex backgrounds and the rounding loss caused by feature map downsampling. A pure analytic geometry method is proposed to refine the keypoints and construct oriented bounding boxes under the criterion of minimum confidence-sensitive offset for keypoints. First, the center is refined by weighted-averaging the coordinates of itself and the shifted positions of its matched midpoints

$$x_c^* = \sum_{i\in\{c,l,t,r,b\}} \frac{w_i \cdot \hat{x}_i'}{5}, \quad y_c^* = \sum_{i\in\{c,l,t,r,b\}} \frac{w_i \cdot \hat{y}_i'}{5} \quad (7)$$

where $w_i = \hat{s}_i / \sum_{j\in\{c,l,t,r,b\}} \hat{s}_j$ is the normalized confidence weight for the center and the shifted midpoints.

Based on the refined centers $(x_c^*, y_c^*)$, the direction of the rotation bounding box with the minimum deviation from the predicted midpoints is formulated and obtained. Fig. 5 illustrates the geometric representation of oriented object elements. According to the geometric characteristics of the rectangle, the connecting lines of symmetric midpoints form two symmetric axes of the rectangle. These two symmetric axes intersect the center and are perpendicular to each other. Thus, the axes can be expressed as

$$\begin{aligned} L_1 &: Y_1 - y_c^* = k\big(X_1 - x_c^*\big), \\ L_2 &: Y_2 - y_c^* = -\frac{1}{k}\big(X_2 - x_c^*\big). \end{aligned} \quad (8)$$

The deviation distance from the midpoint to its closest symmetric axis can be obtained by

$$\begin{aligned} d_i &= \frac{|k\hat{x}_i' - \hat{y}_i' - kx_c^* + y_c^*|}{\sqrt{k^2+1}} \quad \forall i \in \{l, r\}, \\ d_i &= \frac{\left|-\frac{1}{k}\hat{x}_i' - \hat{y}_i' + \frac{1}{k}x_c^* + y_c^*\right|}{\sqrt{\frac{1}{k^2}+1}} \quad \forall i \in \{t, b\}. \end{aligned} \quad (9)$$

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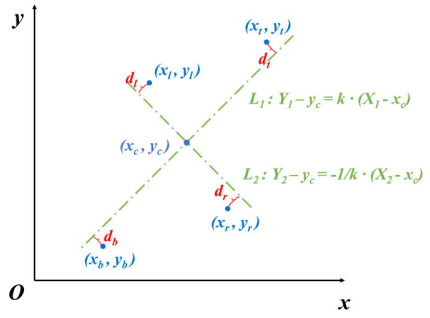



Fig. 5. Mathematical representation of keypoints and symmetric axis in Euclidean coordinate system.

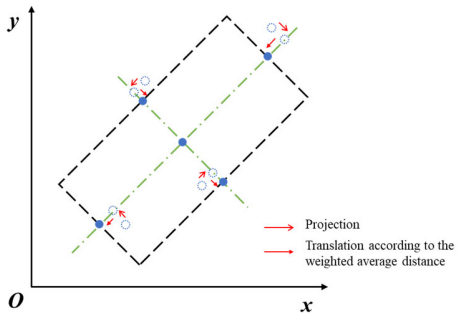



Fig. 6. Midpoints refinement process.

To make the rotation direction consistent with the keypoint estimation as much as possible, the gradient $k^*$ can be solved by minimizing the summed deviation distances

$$k^* = \arg\min_k (d_l + d_t + d_b + d_r). \tag{10}$$

So, the orientation of the object can be obtained from the gradient $k^*$ without unstable angle regression.

Finally, project the midpoint on its nearest symmetric axis, calculate the distance from the midpoint to the center, and refine the midpoint along the symmetric axis based on the distance from the center and its confidence-weighted value, which is illustrated in Fig. 6. In this case, a unique oriented rectangular box can be obtained according to the keypoints, and its class confidence comes from the average confidence of the five keypoints.

*3) Boundary Condition:* When the object is horizontal, the object gradient $k^*$ does not exist or is infinity in the direction calculation process. In this case, MidNet designs a simple postprocessing method. Specifically, after the keypoints grouping and suppression, the gradients of the centripetal shifts $\widehat{cs}$ are calculated. If the number of gradients beyond the threshold exceeds one, this bounding box is treated as a horizontal box. The horizontal bounding box can be defined as $(x_{\min}, y_{\min}, x_{\max}, y_{\max})$, in which the coordinates can be calculated in the following easy way:

$$\begin{aligned} &x_{\min} = \min(x_i),\ y_{\min} = \min(y_i) \\ &x_{\max} = \max(x_i),\ y_{\max} = \max(y_i) \quad \forall i \in \{l, t, r, b\}. \end{aligned} \tag{11}$$

Compared with O$^2$-DNet [20], MidNet realizes explicit keypoint feature learning via the classification branch in a bottom-up way and customizes the symmetrical deformable convolution for oriented objects. Moreover, MidNet realizes the refinement of keypoints in the postprocessing, which reduces the estimation deviation.

## IV. Experimental Results and Analysis

### A. Datasets

*1) HRSC2016:* HRSC2016 is an authoritative remote sensing ship detection dataset, containing 1061 images with 2976 instances from six famous ports from Google Earth. The images of this dataset range in size from 300 × 300 to 1500 × 900. This dataset includes 436 images for training, 181 for validation, and 444 for testing.

*2) FGSD2021:* FGSD2021 is another high-resolution satellite ship detection dataset, which is sampled from the open Google Earth. The dataset contains 636 instances with a standardized GSD and 1 m/pixel. In FGSD2021, the width of the image is ranging from 157 to 7789 pixels, while that of the height is from 224 to 6506 pixels. Compared with HRSC2016, FGSD2021 has much larger image size. The FGSD2021 dataset is divided into 424 and 212 for training and testing, respectively. Following [1], all the 20 annotated categories are used to evaluate the model, which are Aircraft carriers (Air), Wasp class (Was), Tarawa class (Tar), Austin class (Aus), Whidbey Island class (Whi), San Antonio class (San), Newport class (New), Ticonderoga class (Tic), Arleigh Burke class (Bur), Perry class (Per), Lewis and Clark class (Lew), Supply class (Sup), Henry J. Kaiser class (Kai), Bob Hope Class (Hop), Mercy class (Mer), Freedom class (Fre), Independence class (Ind), Avenger class (Ave), Submarine (Sub), and Other (Oth).

### B. Experimental Setting

*1) Multitask Training:* The final loss function is

$$L = L_m + \alpha L_c + L_{\text{cs}} + \beta L_r \tag{12}$$

where $\alpha$ and $\beta$ are the weight coefficients. Since the number of midpoints in the objects is 4× than that of the centers, we set $\beta$ to 0.25. To emphasize the features of the central area of the object, we set $\alpha$ to 0.5.

*2) Implementation Details:* We implemented our method in PyTorch 1.8 and Ubuntu 18.04 with 2 NVIDIA RTX 3090 GPUs. Hourglass-104 [29] is used as the backbone, and it constructs a stacked symmetric architecture composed of bottom–up path and top–down path, which shows superiority in multistage feature extraction and integration for keypoint estimation task. The backbone is pretrained on the COCO dataset [30]. We trained 175 epochs with batch sizes of 4 in the training phase. The initial learning rate is set to 0.0001, and it decays ten times at the 50th and 150th epochs. No more data augmentation is implemented except for random horizontal flip and vertical flip on the two datasets.

### C. Evaluation Metrics

All the methods are evaluated under Pascal VOC 2007 metrics, which calculates the mAP in a 11-point interpolation

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TABLE I
DETECTION ACCURACY ON FGSD2021 DATASET

| | Method | Air | Was | Tar | Aus | Whi | San | New | Tic | Bur | Per | Lew | Sup | Kai | Hop | Mer | Fre | Ind | Ave | Sub | Oth | mAP |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Two-Stage | SCRDet | 77.3 | 90.4 | 87.4 | 89.8 | 78.8 | 90.9 | 54.5 | 88.3 | 89.6 | 74.9 | 68.4 | 59.2 | 90.4 | 77.2 | 81.8 | 73.9 | 100 | 43.9 | 43.8 | 57.1 | 75.90 |
| | $R^2$CNN | 89.9 | 80.9 | 80.5 | 79.4 | 87 | 87.8 | 44.2 | 89 | 89.6 | 79.5 | 80.4 | 47.7 | 81.5 | 87.4 | **100** | 82.4 | **100** | 66.4 | 50.9 | 57.2 | 78.09 |
| | Gliding Vertex | 90 | 90.7 | 90.6 | **90.9** | 89.9 | 77.1 | 38.3 | 79.2 | 90.3 | 76.6 | 79.9 | 79.3 | 90.1 | 81.7 | 80.6 | 71.8 | 81.4 | 78.9 | 57.2 | 73.8 | 79.42 |
| | SASM | 98.1 | 90.3 | 76.9 | 87.8 | 84.8 | 82.8 | 81.3 | 89.8 | 90.6 | 84.4 | 42.7 | 53 | 78.7 | 85.8 | 86.3 | 80.7 | 88.1 | 62.6 | 60.9 | 83.2 | 79.44 |
| | Oriented R-CNN | 90.9 | 89.7 | 81.5 | 81.1 | 79.6 | 88.2 | **98.9** | 89.8 | 90.6 | 87.8 | 60.4 | 73.9 | 81.8 | 86.7 | **100** | 60 | **100** | 79.4 | 66.9 | 63.7 | 82.54 |
| | ROI-Trans. | 90.9 | 88.6 | 87.2 | 89.5 | 78.5 | 88.8 | 81.8 | 89.6 | 89.8 | **90.4** | 71.7 | 74.7 | 73.7 | 81.6 | 78.6 | **100** | 75.6 | 78.4 | 68 | 66.9 | 83.48 |
| | CFA | **99.9** | 90.5 | **95.5** | 88.4 | 85.6 | 83.8 | **95.6** | 86.7 | **90.8** | 88.2 | 73.4 | 54 | 89.3 | 81 | 79.3 | **88** | **98.5** | **89** | 58.8 | **88.6** | 85.25 |
| | ReDet | 90.9 | 90.6 | 80.3 | 81.5 | 89.3 | 88.4 | 81.8 | 88.8 | 90.3 | **90.5** | 78.1 | 76 | **90.7** | 87 | **98.2** | 84.4 | 90.9 | 74.6 | **85.3** | 71.2 | 85.44 |
| Single-Stage | GWD | 90.9 | 88.9 | 70.6 | 63.2 | 66.3 | 55.8 | 43.4 | 81.8 | 87.7 | 83.8 | 25.2 | 9.1 | 48.8 | 62.2 | 33.2 | 69.9 | 62.7 | 60.4 | 63.3 | 80.2 | 62.37 |
| | $R^3$Det | 90.9 | 80.9 | 81.5 | **90.1** | 79.3 | 87.5 | 29.5 | 77.4 | 89.4 | 69.7 | 59.9 | 67.3 | 80.7 | 76.8 | 72.7 | 83.3 | 90.9 | 38.4 | 23.1 | 40 | 70.47 |
| | DCL | 89.9 | 81.4 | 78.6 | 80.7 | 78 | 87.9 | 49.8 | 78.7 | 87.2 | 76.1 | 60.6 | 76.9 | **90.4** | 80 | 78.8 | 77.9 | **100** | 37.1 | 31.2 | 45.6 | 73.34 |
| | RetinaNet-O | 89.7 | 89.2 | 78.2 | 87.3 | 77 | 86.9 | 62.7 | 81.5 | 83.3 | 70.6 | 46.8 | 69.9 | 80.2 | 83.1 | **100** | 80.6 | 89.7 | 61.5 | 42.5 | 9.1 | 73.49 |
| | CSL | 89.7 | 81.3 | 77.2 | 80.2 | 71.4 | 77.2 | 52.7 | 87.7 | 87.7 | 74.2 | 57.1 | **97.2** | 77.6 | 80.5 | **100** | 72.7 | **100** | 32.6 | 37 | 40.7 | 73.73 |
| | RSDet | 89.8 | 80.4 | 75.8 | 77.3 | 78.6 | 88.8 | 26.1 | 84.7 | 87.6 | 75.2 | 55.1 | 74.4 | 89.7 | 89.3 | **100** | 86.4 | **100** | 27.6 | 37.6 | 50.6 | 73.74 |
| | $S^2$A-Net | 90.9 | 81.4 | 73.3 | 89.1 | 80.9 | 89.9 | 81.2 | 89.2 | 90.7 | 88.9 | 60.5 | 75.9 | 81.6 | 89.2 | **100** | 68.6 | 90.9 | 61.3 | 55.7 | 64.7 | 80.19 |
| | DARDet | 90.9 | 89.2 | 69.7 | 89.6 | 88 | 81.4 | 90.3 | 89.5 | 90.5 | 79.7 | 62.5 | **87.9** | 90.2 | 89.2 | **100** | 68.9 | 81.8 | 66.3 | 44.3 | 56.2 | 80.31 |
| | KFIoU | **100** | **99.8** | 90.9 | **90.9** | **90.8** | 89.6 | 80.1 | **90.7** | **90.9** | 87.3 | **82.8** | 66.7 | **90.4** | 90.2 | 89.7 | 86.5 | 90.8 | 46.4 | 69.2 | **90.1** | 85.69 |
| | KLD | **100** | **90.9** | **96** | **90.9** | **90.9** | **90.5** | 81.4 | **90.9** | **90.9** | 88.2 | **85** | 66.7 | **90.7** | **90.6** | 97.2 | 84.7 | 94.2 | 41.7 | 68.4 | 88.7 | **85.93** |
| | MidNet (Our) | 90.9 | 88.6 | 79 | 89 | 83.2 | 89.2 | **98.9** | 88.9 | 90.3 | 90.1 | 62.5 | 82.4 | 89.8 | **95.8** | 95.5 | 87.6 | 87 | **84.5** | **83.9** | 73.1 | **86.50** |

fashion. Specifically, for each category of object, the precision $P$ and recall $R$ are defined as

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}$$
$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{13}$$

where the TP, FP, and FN refer to the numbers of true positives, false positives, and false negatives, respectively.

The AP is summarized by averaging the maximum precision values at a set of 11 equally spaced recall levels [0, 0.1, 0.2, . . . ,1], which is given by

$$\mathrm{AP} = \frac{1}{11} \sum_{R \in \{0, 0.1, \ldots, 0.9, 1\}} P_{\mathrm{interp}}(R) \tag{14}$$

where

$$P_{\mathrm{interp}}(R) = \max_{\tilde{R}:\tilde{R}>R} P\left(\tilde{R}\right). \tag{15}$$

The mAP is the average AP over all the classes, which is

$$\mathrm{mAP} = \frac{1}{N_{\mathrm{cls}}} \sum_{i=1}^{N} \mathrm{AP}_i \tag{16}$$

where $\mathrm{AP}_i$ is the AP in the $i$th class and $N_{\mathrm{cls}}$ is the total number of classes being evaluated.

TABLE II
DETECTION ACCURACY ON HRSC2016 DATASET

| Type | Method | Backone | Input_size | AP |
|---|---|---|---|---|
| Two-Stage | BL2 | VGG16 | - | 69.60 |
| | $R^2$CNN | ResNet101 | 800×800 | 73.07 |
| | RoI-Transformer | ResNet101 | 512×800 | 86.2 |
| | Gliding Vertex | ResNet101 | 512×800 | 88.2 |
| | SASM | ResNet101 | 512×800 | 90.27 |
| | ReDet | ResNet101 | 512×800 | 90.46 |
| | Oriented R-CNN | ResNet101 | 1333×800 | **90.5** |
| Single-Stage | RSDet | ResNet50 | 800×800 | 86.5 |
| | CPS-Det | ResNeXt50 | 800×800 | 89.12 |
| | RetinaNet-R | ResNet101 | 800×800 | 89.18 |
| | $R^3$Det | ResNet101 | 800×800 | 89.26 |
| | KFIOU | ResNet50 | 500×500 | 89.43 |
| | DCL | ResNet101 | 800×800 | 89.46 |
| | GRS-Det | ResNet101 | 800×800 | 89.57 |
| | $AR^2$Det | ResNet34 | 512×512 | 89.58 |
| | CSL | ResNet50 | 800×800 | 89.62 |
| | GWD | ResNet101 | 800×800 | 89.85 |
| | KLD | ResNet50 | 500×500 | 89.97 |
| | $S^2$A-Net | ResNet101 | 512×800 | 90.17 |
| | DARDet | Resnet50 | 1024×1024 | 90.37 |
| | MidNet (Our) | Hourglass104 | 512×512 | **90.52** |

### D. Comparison With the State-of-the-Art Methods

We compare our proposed method with the state-of-the-art methods (SCRDet [16], $R^2$CNN [31], gliding vertex [4], SASM [32], oriented R-CNN [5], ROI-Trans. [12], CFA [33], ReDet [13], GWD [34], $R^3$Det [14], DCL [19], RetinaNet-O [35], CSL [18], RSDet [17], $S^2$A-Net [15], DARDet [36], KFIoU [37], KLD [38], CPS-Det [26], GRS-Det [2], and $AR^2$Det [3]) on the two datasets, and Tables I and II report the detailed comparison results. For fair comparison, CHPDet [1] is not considered as a comparison because it introduces additional annotation information. The results of comparison algorithms are obtained by [1] or the implementation in mmrotate [35], with the default configuration.

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TABLE III
DETECTION ACCURACY ON FGSD2021 DATASET AT DIFFERENT IOU THRESHOLDS

| Method | | $mAP_{0.5}$ | $mAP_{0.6}$ | $mAP_{0.7}$ | $mAP_{0.8}$ |
|---|---|---|---|---|---|
| Two -Stage | SCRDet | 75.90 | 70.98 | 61.82 | 35.12 |
| | $R^2$CNN | 78.09 | 75.03 | 64.83 | 36.41 |
| | Gliding Vertex | 79.42 | 77.45 | 70.85 | 57.45 |
| | SASM | 79.44 | 74.29 | 68.10 | 48.54 |
| | Oriented R-CNN | 82.54 | 81.32 | 78.53 | 64.87 |
| | ROI-Trans. | 83.48 | 82.63 | 80.35 | 65.18 |
| | CFA | 85.25 | 84.44 | 78.23 | **68.90** |
| | ReDet | 85.44 | **84.65** | **80.24** | 67.94 |
| Single -Stage | GWD | 62.37 | 58.76 | 55.37 | 38.17 |
| | $R^3$Det | 70.47 | 68.32 | 57.17 | 27.44 |
| | DCL | 73.34 | 69.19 | 57.80 | 28.54 |
| | RetinaNet-O | 73.49 | 69.17 | 62.82 | 45.55 |
| | CSL | 73.73 | 69.71 | 60.25 | 34.93 |
| | RSDet | 73.74 | 69.55 | 61.52 | 35.83 |
| | $S^2$A-Net | 80.19 | 79.58 | 75.65 | 58.82 |
| | DARDet | 80.31 | 79.62 | 74.77 | 59.21 |
| | KFIoU | 85.69 | 83.65 | 77.22 | 56.50 |
| | KLD | **85.93** | 83.53 | 79.24 | 60.66 |
| | MidNet (Our) | **86.50** | **85.36** | **81.33** | **68.56** |

For the FGSD2012 dataset, 20 categories are used to evaluate the detectors, and the results are recoded in Table I. MidNet achieves an mAP of 86.5% and improves by 1.06% and 0.57% compared with the ReDet and KLD, which are the best performance among two- and single-stage detectors. For the anchor-free methods, MidNet outperforms the state-of-the-art (DARDet) 6.29% for mAP. MidNet achieves the best performance among all the detectors for Newport class and Bob Hope class and the second performance in Avenger class and Submarine. This proves that the MidNet has advantages in the multiclassification task. Moreover, the detection performance at different IoU thresholds displayed in Table III obviously shows that MidNet achieves the state-of-the-art under various thresholds, and the mAP still remained at 68.56% even if the IoU threshold was harshly set to 0.8, which clarifies the superiority of the keypoint-based representation for oriented objects.

For the HRSC2016 dataset, the overall comparison performance is reported in Table II. Our proposed method achieves 90.52% AP and outperforms all the existing detectors. The performance gain is 20.92% compared with the official baseline of HRSC2016 (BL2 [22]), which indicates the significant advantage of MidNet. Compared with the state-of-the-art methods of two-stage (oriented R-CNN) and single-stage (DARDet), MidNet outperforms them by 0.02% and 0.15%. For the ship specific detectors (CPS-Det, GRS-Det, and AR2Det), MidNet achieves an increase of 1.4%, 0.95%, and 0.94% of AP, respectively.

### E. Ablation Studies

To check the effectiveness of each part in MidNet, the ablation experiment is implemented. Table IV verifies each module in the network, and Table V investigates the effectiveness of the postprocessing method.

TABLE IV
ABLATION EXPERIMENTS FOR EACH MODULE IN MIDNET

| Corners Prediction | Midpoints Prediction | Center Prediction | Symmetrical DCN | FGSD 2021 | HRSC 2016 |
|---|---|---|---|---|---|
| ✓ | | | | 80.09 | 86.24 |
| | ✓ | | | 83.17 | 88.87 |
| | ✓ | ✓ | | 85.23 | 89.65 |
| | ✓ | ✓ | ✓ | **86.50** | **90.52** |

TABLE V
ABLATION EXPERIMENTS FOR POSTPROCESSING METHOD

| Method | FGSD 2021 | HRSC 2016 |
|---|---|---|
| Keypoint matching based on ExtremeNet pipeline | 83.76 | 90.04 |
| Construct bounding box without gradient calculation | 85.12 | 90.45 |
| Our method | **86.50** | **90.52** |

*1) Corners Versus Midpoints:* To verify the superiority of midpoint representation, we use MidNet to predict the corners for comparison in Table IV. Due to the uncertainty of corner distribution, we remove the corner pooling to avoid chaotic feature learning. Experiments show that without any additional structure, midpoint-based feature learning improves the AP by 3.08% and 2.63% on the two datasets.

*2) Center Point Prediction:* In the aerial images, the size of ships varies greatly, and there are many long and narrow ships. If the manually set matching range is too large, it may cause incorrect matching of keypoints with low confidence. If the range is too small, it may cause missed detection. MidNet adaptively delimits the matching range by predicting the center point and its matching radius. Experiment shows that the midpoint-based model improved the AP by 2.06% and 0.78% through incorporating the center prediction.

The results of two different keypoint matching strategies on the HRSC 2016 dataset are shown in Fig. 9. The blue bounding box is the prediction result, the green bounding box is ground truth, and the yellow dot is the reserved keypoint. Fig. 9(a) and (b) shows the keypoints matched by manually set range and by the proposed adaptive method. If a large range value is set, the manually set manner causes incorrect matching for long, narrow, and side-by-side ships in Fig. 9(a) (left). On the contrary, a small range value leads to missed detection of the large-sized ship in Fig. 9(a) (right).

*3) Symmetrical Deformable Convolution:* The experimental results show that the detection accuracy is improved by 1.27% and 0.87% by using effective symmetrical deformable convolution. By sharing the offset, the optimization of the feature map and offset promotes each other, which is the key for MidNet to locate the keypoints accurately.

*4) Postprocessing Method:* To check the effectiveness of the proposed keypoint matching strategy, we implement the ExtremeNet pipeline on MidNet. In ExtremeNet, any valid combination of keypoints is retained to build a bounding box,

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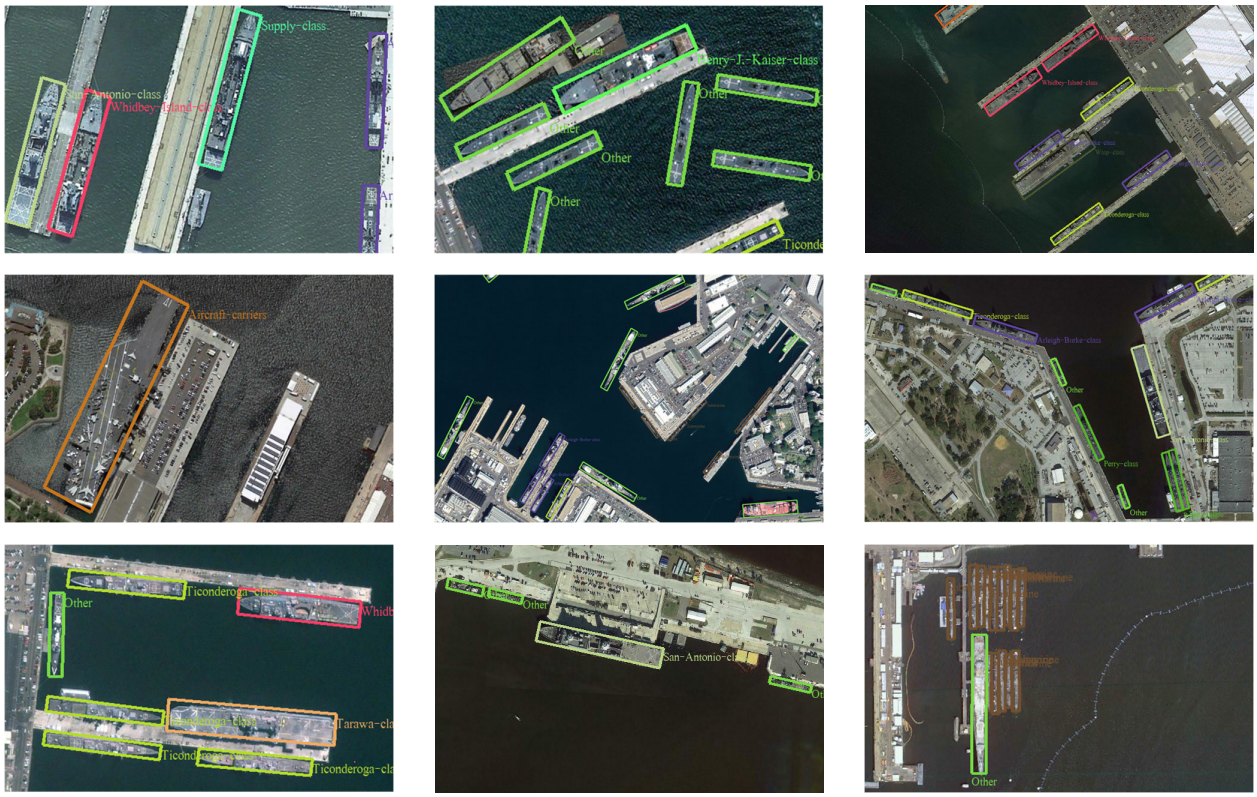



Fig. 7. Visual detection results on the FGSD2021 dataset.

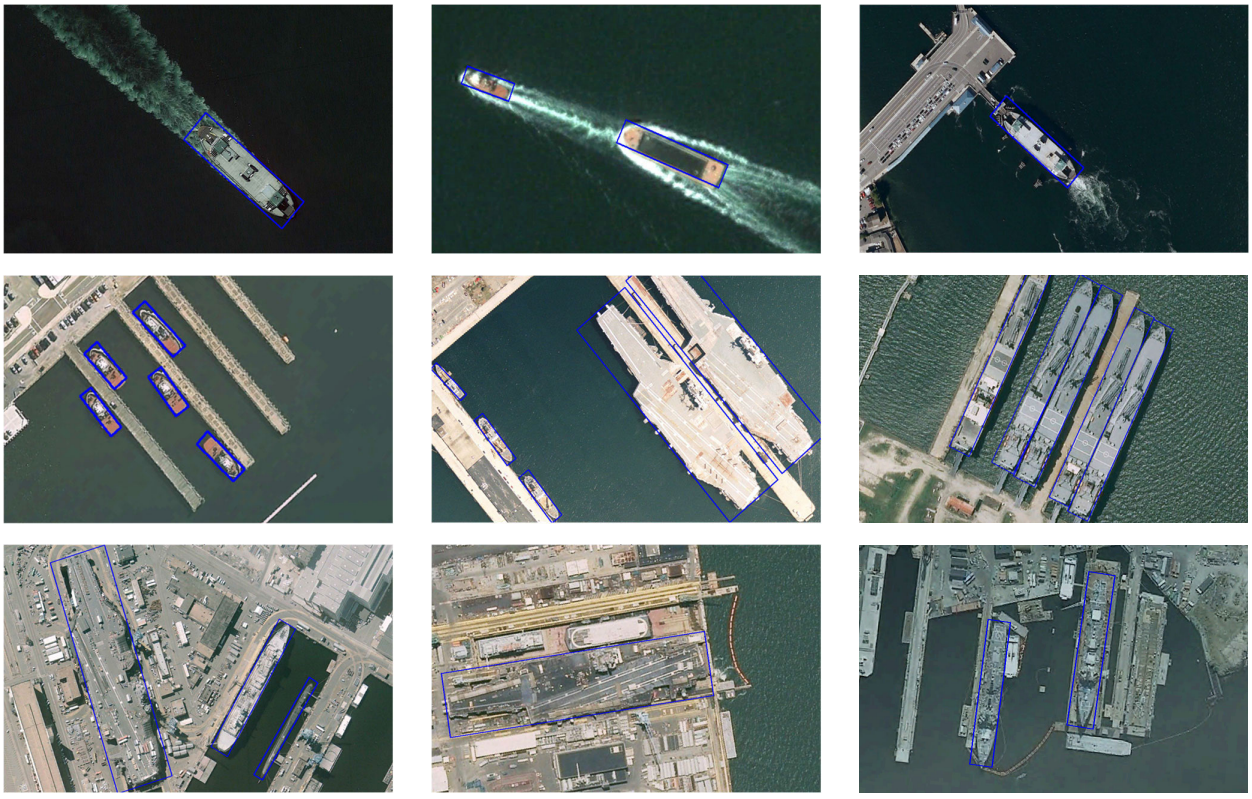


Fig. 8. Visual detection results on the HRSC2016 dataset.

and then, NMS is carried out in the whole image. As shown in Table V, this exhaustive strategy reduces the AP by 2.74% and 0.48%, which is caused by high-confidence incorrect keypoint matchings. In addition, the time complexity of this method

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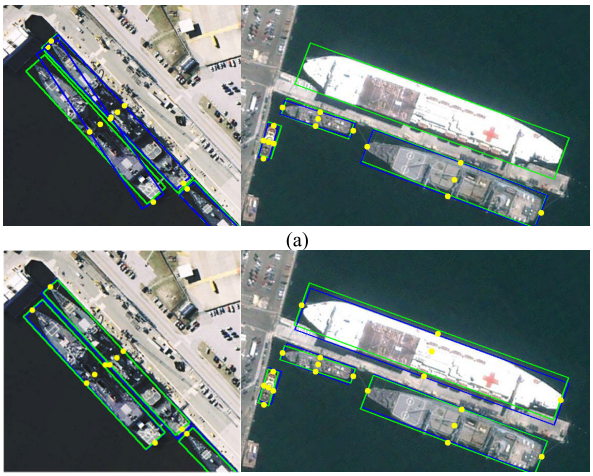


Fig. 9. Results of different keypoint matching strategies on the HRSC2016 dataset. (a) and (b) Keypoints matched by manually set range and by the proposed adaptive method, respectively.

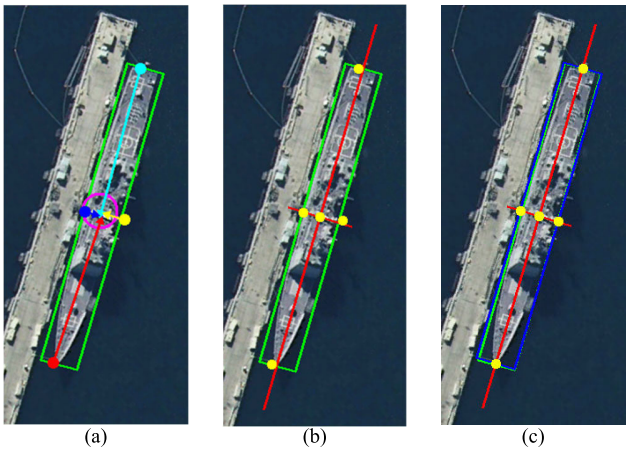


Fig. 10. Visualization of postprocessing on the HRSC2016 dataset. (a) Predicted keypoints, centripetal shifts and matching vectors. (b) Calculated symmetric axes. (c) Adjusted keypoints and the final detection result.

reaches $O(n^4)$, which is unacceptable in the crowded remote sensing scene. Moreover, a simplified algorithm for bounding box construction without gradient calculation is designed for comparison. Specifically, connect the symmetrical midpoints and then translate the connection until it passes through the remaining two midpoints and get the two sides of the bounding box. Repeat this strategy at another pair of midpoints to get the remaining two sides, so the bounding box is generated. The mAP decreases 1.38% and 0.07% without angle calculation or any keypoint refinement, but the calculated quantities reduce partly meanwhile. Therefore, this method can also be regarded as a simplified alternative.

The visualization of general postprocessing on the HRSC 2016 dataset is shown in Fig. 10. Fig. 10(a) shows the predicted keypoints, centripetal shifts, and matching vectors, Fig. 10(b) shows the two calculated symmetric axes $L_1$ and $L_2$, and Fig. 10(c) shows the adjusted keypoint and the final detection result. In Fig. 10, the green box is ground truth, and the predicted result is marked in blue. In Fig. 10(a), MidNet matches the keypoints by using centripetal shifts and matching vector correctly, and it can be seen in Fig. 10(b) that the symmetric axes can be located. Based on the calculated symmetric axes and refined keypoints, a precise oriented bounding box is generated in Fig. 10(c).

TABLE VI
DETECTION RESULTS OF MIDNET WITH DIFFERENT BACKBONES

| | ResNet -101-FPN | ResNeXt -101-FPN | DLA -34 | Hourglass -52 | Hourglass -104 |
|---|---|---|---|---|---|
| FGSD2021 | 84.83 | 85.57 | 84.49 | 84.33 | **86.50** |
| HRSC2016 | 89.76 | 90.03 | 88.61 | 89.80 | **90.52** |

TABLE VII
DETECTION ACCURACY OF DIVERSE OBJECT ENCODING MANNERS BASED ON RESNET-101

| Method | Backbone | FGSD2021 | HRSC2016 | Coding Manner |
|---|---|---|---|---|
| ROI-Trans. | ResNet-101 | 84.36 | 86.20 | Oriented anchors |
| Gliding Vertex | ResNet-101 | 83.08 | 88.20 | Horizontal anchors |
| $O^2$-DNet | ResNet-101 | 84.55 | 88.62 | Middle line |
| MidNet-corners | ResNet-101 | 82.84 | 85.32 | Corner point |
| MidNet | ResNet-101 | **84.83** | **89.76** | Midpoint & Center |

TABLE VIII
DETECTION RESULTS OF MIDNET WITH DIFFERENT λ VALUES

| | 1.0 | 0.9 | 0.8 | 0.7 | 0.6 | 0.5 | 0.4 |
|---|---|---|---|---|---|---|---|
| FGSD2021 | 84.66 | 84.06 | 85.17 | 85.04 | **86.50** | 85.67 | 83.87 |
| HRSC2016 | 88.09 | 89.44 | 89.92 | 90.34 | **90.52** | 89.34 | 87.01 |

### *F. Sensitivity Analysis of Different Architectures and Parameters*

*1) Backbones:* MidNet is implemented with five different architectures: ResNet-101 [39], ResNeXt-101 [40], DLA-34 [41], Hourglass-52, and Hourglass-104.

It can be seen from Table VI that MidNet gains competitive performance on the FGSD2021 and HRSC2016 datasets with different backbone networks. Hourglass-104 shows improved performance when MidNet is equipped with 100-layer scale backbones. Without implementing cascading strategies, the advantage of Hourglass-52 is diminished for lightweight networks. In the FGSD2021 dataset, Hourglass-52's performance is slightly inferior to that of the DLA-34, although the difference is minimal. Considering that there is only a minimal decline in accuracy when equipped with the most lightweight backbone, we think that MidNet displays robust performance despite changes to the backbone architecture.

*2) Coding Manners:* To confirm the superiority of midpoint representation, we tested the performance of existing oriented object encoding manners and our proposed MidNet with fixed backbones. The experimental results are presented in Table VII. MidNet-corners refer to MidNet based on corner prediction. As shown in Table VII, MidNet achieves superior performance when using ResNet-101 as the backbone. Regarding ROI-Trans and Gliding Vertex, the hyperparameters of the anchors limit the model's generalization in complex

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TABLE IX
DETECTION ACCURACY ON THE DOTA DATASET

| | Method | Backbone | PL | BD | BR | GTF | SV | LV | SH | TC | BC | ST | SBF | RA | HA | SP | HC | mAP |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Two-stage | Faster RCNN-O | R-101 | 79.09 | 69.12 | 17.17 | 63.49 | 34.20 | 37.16 | 36.20 | 89.19 | 69.60 | 58.96 | 49.40 | 52.52 | 46.69 | 44.80 | 46.30 | 52.93 |
| | ROI-Trans | R-101 | 88.64 | 78.52 | 43.44 | 75.92 | 68.81 | 73.68 | 83.59 | 90.74 | 77.27 | 81.46 | 58.39 | 53.54 | 62.83 | 58.93 | 47.67 | 69.56 |
| | SCRDet | R-101 | 89.98 | 80.65 | 52.09 | 68.36 | 68.36 | 60.32 | 72.41 | 90.85 | **87.94** | **86.86** | 65.02 | 66.68 | 66.25 | 68.24 | 65.21 | 72.61 |
| | CFA | R-101 | 89.26 | 81.72 | 51.81 | 67.17 | 79.99 | 78.25 | 84.46 | 90.77 | 83.40 | 85.54 | 54.86 | 67.75 | 73.04 | 70.24 | 64.96 | 74.88 |
| | SASM | R-50 | 86.42 | 78.97 | 52.47 | 69.84 | 77.30 | 75.99 | 86.72 | 90.89 | 82.63 | 85.66 | 60.13 | 68.25 | 73.98 | 72.22 | 62.37 | 74.92 |
| | Gliding Vertex | R-101 | 89.64 | **85.00** | 52.26 | **77.34** | 73.01 | 73.14 | 86.82 | 90.74 | 79.02 | 86.81 | 59.55 | **70.91** | 72.94 | 70.86 | 57.32 | 75.02 |
| | ReDet | ReR-50 | 88.79 | 82.64 | 53.97 | 74.00 | 78.13 | **84.06** | 88.04 | 90.89 | 87.78 | 85.75 | 61.76 | 60.39 | 75.96 | 68.07 | 63.59 | 76.25 |
| | Oriented R-CNN | R-101 | 88.86 | 83.48 | 55.27 | **76.92** | 74.27 | 82.10 | 87.52 | **90.90** | 85.56 | 85.33 | **65.51** | 66.82 | 74.36 | 70.15 | 57.28 | 76.29 |
| One-stage | GWD | R-50 | 89.09 | 79.22 | 45.78 | 67.94 | 66.58 | 56.29 | 71.55 | 90.80 | 80.50 | 76.61 | 58.53 | 61.01 | 59.92 | 69.51 | 57.59 | 68.73 |
| | RetinaNet-O | R-101 | 88.67 | 77.62 | 41.81 | 58.17 | 74.58 | 71.64 | 79.11 | 90.29 | 82.18 | 74.32 | 54.75 | 60.60 | 62.57 | 69.67 | 60.64 | 69.77 |
| | DCL | R-152 | **90.20** | 83.50 | 53.60 | 70.10 | 64.60 | 79.40 | 67.30 | **91.00** | **88.30** | 82.50 | 64.00 | **68.70** | 62.80 | 69.50 | 66.90 | 73.49 |
| | CSL | R-152 | 88.76 | 83.09 | 50.91 | 67.27 | 76.23 | 80.39 | 86.72 | 90.78 | 84.68 | 83.24 | 61.98 | 61.35 | 66.91 | 70.63 | 53.94 | 73.79 |
| | RSDet | R-50 | 89.11 | 82.84 | 48.37 | 71.11 | 78.11 | 78.39 | 87.25 | 90.83 | 84.90 | 85.64 | 60.36 | 62.60 | 65.26 | 69.13 | 57.94 | 74.12 |
| | $S^2$A-Net | R-152 | 89.78 | 83.95 | 52.63 | 69.70 | 76.84 | 81.26 | 87.30 | 90.81 | 84.67 | 85.27 | 63.50 | 64.16 | 68.96 | 68.79 | 65.45 | 75.54 |
| | DARDet | R-50 | 88.82 | 82.94 | **55.63** | 72.75 | 78.52 | 83.10 | 87.46 | 90.21 | 86.36 | 85.44 | 64.70 | 61.41 | 73.46 | 76.94 | 57.38 | 76.34 |
| | KFIoU | R-50 | 89.04 | 84.04 | 52.98 | 73.00 | 78.69 | 83.60 | 87.61 | 90.79 | 85.97 | 85.47 | 64.77 | 63.29 | 69.18 | 76.38 | 65.63 | 76.70 |
| | KLD | R-50 | 88.90 | 84.17 | **55.80** | 69.35 | 78.72 | **84.08** | 87.00 | 89.75 | 84.23 | 85.73 | 64.74 | 61.80 | **76.72** | **78.49** | **70.89** | **77.36** |
| | $R^3$Det | R-50 | 88.89 | **84.31** | 55.32 | 75.49 | **80.33** | 81.69 | **88.24** | 90.88 | 83.62 | **87.46** | 59.85 | 65.60 | **76.86** | **80.46** | 65.17 | **77.61** |
| | MidNet (Our) | H-104 | **89.99** | 84.28 | 50.06 | 75.02 | **80.27** | 82.54 | **88.65** | 90.25 | 86.42 | 86.20 | **66.81** | 67.60 | 69.18 | 64.30 | **68.29** | 76.66 |



Fig. 11. Detection results on DOTA dataset.

scenarios. $O^2$-DNet [20] employs an implicit regression to determine keypoint positions. Meanwhile, the absence of fine-tuned postprocessing and the utilization of structural priors limit its accuracy of object localization furtherly. Corner prediction introduces ambiguity in the definition of corners due to arbitrary angles of objects, resulting in difficulty in feature learning. Experimental results demonstrate that the midpoint encoding manner performs better on oriented ships.

*3) Matching Radius:* Matching radius has an important impact on keypoint matching. We try to change the scaling factor $\lambda$ to scale the matching radius, and the results are recorded in Table VIII. Due to the nonnegligible prediction error, too large matching radius is possible to produce the prediction beyond the object, especially for the ships with large aspect ratio. In this case, when the objects are densely arranged, it is easy to generate incorrect keypoint matching. For too small matching radius, the matching area of the center

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is shrunk. When the centripetal vector deviates, it is likely to miss the midpoint. As shown in Table VIII, 0.6 is the most appropriate scaling ratio for both datasets. Significantly, this parameter is different from the manually set matching radius. The matching radius varies with various object sizes, so it is difficult to determine for different objects. The parameter $\lambda$ is a scaling factor, which is robust and easy to determine for objects.

### G. Generality Ability Verification

To verify the generality of the proposed method, we evaluate the performance of MidNet on various categories of aerial objects on the DOTA dataset. The DOTA dataset is comprised of 2806 large aerial images from different sensors and platforms and contains 188 282 instances in 15 categories: plane (PL), baseball diamond (BD), bridge (BR), ground field track (GTF), small vehicle (SV), large vehicle (LV), SH-ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer-ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP), and helicopter (HC). Objects in DOTA exhibit a wide variety of scales, orientations, and shapes. The original images ranging from $800 \times 800$ to $1500 \times 900$ pixels, which are cropped into $800 \times 800$ patches with a stride of 600.

Following the previous works, we use training and validation sets for training and the rest for testing. Random rotation is adopted to avoid overfitting during the training. We train MidNet with 50 epochs. SGD optimizer is adopted with an initial learning rate of 0.01, which decays ten times at the 15th and 35th epochs. The detection results of all the methods are shown in Table IX. In Table IX, R-50, R-101, and R-152 stand for ResNet-50, ResNet-101, and ResNet-152, respectively. ReR-50 is ReResNet-50, H-104 is the 104-layer hourglass network, and DLA-34 means the 34-layer deep layer aggregation network.

As shown in Table IX, MidNet achieves a competitive result of 76.66% mAP on the DOTA dataset, which is 0.95% lower than the state-of-the-art detector ($R^3$Det). For the approximately symmetric objects, MidNet achieves excellent results. For ship and soccer field objects, MidNet obtains the highest AP, and for plane and small-vehicle, MidNet yields the second best with a weak disadvantage. However, for some extremely asymmetric objects, such as swimming pool and harbor, the detection accuracy still needs to be further improved. The experimental results conform to the design concept of MidNet. Fig. 11 shows the visual detection results on the DOTA dataset.

## V. Conclusion

We present a novel midpoints-based oriented ship detection framework. It predicts the position and matching vector of five keypoints (midpoints and center) and groups these keypoints to construct the oriented bounding box in a purely analytical geometry manner. To mine the discriminative features of keypoints in the complex background, a symmetrical deformable convolution is proposed based on the structural characteristics of ships. Our method provides a new perspective for predicting the oriented object by means of keypoints, which avoids unstable angle prediction and overparameterized anchor mechanism. The state-of-the-art detection results on HRSC2016 and FGSD2021 demonstrate the superiority of the proposed framework.

## References


[1] F. Zhang, X. Wang, S. Zhou, Y. Wang, and Y. Hou, "Arbitrary-oriented ship detection through center-head point extraction," 2021, *arXiv:2101.11189*.

[2] X. Zhang, G. Wang, P. Zhu, T. Zhang, C. Li, and L. Jiao, "GRS-Det: An anchor-free rotation ship detector based on Gaussian-mask in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 4, pp. 3518–3531, Apr. 2021.

[3] Y. Yang et al., "$AR^2$Det: An accurate and real-time rotational one-stage ship detector in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5605414.

[4] Y. Xu et al., "Gliding vertex on the horizontal bounding box for multi-oriented object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1452–1459, Apr. 2021.

[5] X. Xie, G. Cheng, J. Wang, X. Yao, and J. Han, "Oriented R-CNN for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 3500–3509.

[6] Z. Yang, S. Liu, H. Hu, L. Wang, and S. Lin, "RepPoints: Point set representation for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9656–9665.

[7] X. Zhou, J. Zhuo, and P. Krähenbuhl, "Bottom-up object detection by grouping extreme and center points," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 850–859.

[8] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 734–750.

[9] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6568–6577.

[10] Z. Dong, G. Li, Y. Liao, F. Wang, P. Ren, and C. Qian, "CentripetalNet: Pursuing high-quality keypoint pairs for object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 10516–10525.

[11] J. Ma et al., "Arbitrary-oriented scene text detection via rotation proposals," *IEEE Trans. Multimedia*, vol. 20, no. 11, pp. 3111–3122, Nov. 2018.

[12] J. Ding, N. Xue, Y. Long, G. Xia, and Q. Lu, "Learning RoI transformer for oriented object detection in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2844–2853.

[13] J. Han, J. Ding, N. Xue, and G. Xia, "ReDet: A rotation-equivariant detector for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2785–2794.

[14] X. Yang, J. Yan, Z. Feng, and T. He, "$R^3$Det: Refined single-stage detector with feature refinement for rotating object," 2019, *arXiv:1908.05612*.

[15] J. Han, J. Ding, J. Li, and G.-S. Xia, "Align deep features for oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5602511.

[16] X. Yang et al., "SCRDet: Towards more robust detection for small, cluttered and rotated objects," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8231–8240.

[17] W. Qian, X. Yang, S. Peng, J. Yan, and Y. Guo, "Learning modulated loss for rotated object detection," in *Proc. AAAI Conf. Arti. Intell.*, May 2021, vol. 35, no. 3, pp. 2458–2466.

[18] X. Yang and J. Yan, "On the arbitrary-oriented object detection: Classification based approaches revisited," *Int. J. Comput. Vis.*, vol. 130, no. 5, pp. 1340–1365, 2022.

[19] X. Yang, L. Hou, Y. Zhou, W. Wang, and J. Yan, "Dense label encoding for boundary discontinuity free rotation detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 15814–15824.

[20] H. Wei, Y. Zhang, Z. Chang, H. Li, H. Wang, and X. Sun, "Oriented objects as pairs of middle lines," *ISPRS J. Photogramm. Remote Sens.*, vol. 169, pp. 268–279, Nov. 2020.

[21] J. Dai et al., "Deformable convolutional networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 764–773.

[22] Z. Liu, L. Yuan, L. Weng, and Y. Yang, "A high resolution optical satellite image dataset for ship recognition and some new baselines," in *Proc. 6th Int. Conf. Pattern Recognit. Appl. Methods*, 2017, pp. 324–331.


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[23] Z. Liu, J. Hu, L. Weng, and Y. Yang, "Rotated region based CNN for ship detection," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Sep. 2017, pp. 900–904.
[24] T. Wang and Y. Li, "Rotation-invariant task-aware spatial disentanglement in rotated ship detection based on the three-stage method," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5609112.
[25] H. Guo, N. Wang, B. Song, and X. Gao, "A rotational libra R-CNN method for ship detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 8, pp. 5772–5781, Aug. 2020.
[26] X. Li et al., "Ship detection of optical remote sensing image in multiple scenes," *Int. J. Remote Sens.*, vol. 43, pp. 5709–5737, Aug. 2021.
[27] M. Zhu et al., "Arbitrary-oriented ship detection based on RetinaNet for remote sensing images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 6694–6706, 2021.
[28] T. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.
[29] T. Xu and W. Takano, "Graph stacked hourglass networks for 3D human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 483–499.
[30] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. Eur. Conf. Comput. Vis.*, 2014, pp. 740–755.
[31] Y. Jiang et al., "$R^2$ CNN: Rotational region CNN for arbitrarily-oriented scene text detection," in *Proc. 24th Int. Conf. Pattern Recognit. (ICPR)*, Aug. 2018, pp. 3610–3615.
[32] L. Hou, K. Lu, J. Xue, and Y. Li, "Shape-adaptive selection and measurement for oriented object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2022, pp. 923–932.
[33] Z. Guo, C. Liu, X. Zhang, J. Jiao, X. Ji, and Q. Ye, "Beyond bounding-box: Convex-hull feature adaptation for oriented and densely packed object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8788–8797.
[34] X. Yang, J. Yan, Q. Ming, W. Wang, X. Zhang, and Q. Tian, "Rethinking rotated object detection with Gaussian Wasserstein distance loss," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 11830–11841.
[35] Y. Zhou et al., "MMRotate: A rotated object detection benchmark using PyTorch," 2022, *arXiv:2204.13317*.
[36] F. Zhang, X. Wang, S. Zhou, and Y. Wang, "DARDet: A dense anchor-free rotated object detector in aerial images," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[37] X. Yang et al., "The KFIoU loss for rotated object detection," 2022, *arXiv:2201.12558*.
[38] X. Yang et al., "Learning high-precision bounding box for rotated object detection via Kullback–Leibler divergence," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 18381–18394.
[39] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[40] S. Xie, R. Girshick, P. Dollár, Z. Tu, and K. He, "Aggregated residual transformations for deep neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5987–5995.
[41] F. Yu, D. Wang, E. Shelhamer, and T. Darrell, "Deep layer aggregation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 2403–2412.



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